ONLINE SUPPLEMENTARY MATERIAL

A Single High-Fat Meal Adversely Affects Postprandial Vascular Endothelial Function: A Systematic Review and Meta-Analysis of Clinical Trials.

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Supplementary Table 1. Full search strategy and search terms in Ovid MEDLINE

Ovid MEDLINE(R) and Epub Ahead of Print, In-Process, In-Data-Review & Other Non-Indexed Citations, Daily and Versions(R): 1946 to 20 January 2022
1. high adj2 fat*.mp
2. High fat*.mp
3. oral fat load.mp
4. lipid*.mp
5. triglyceride*.mp
6. dietary fat*.mp
7. MeSH exp Lipids/
8. MeSH exp Dietary Fats/
9. MeSH Diet, High-Fat/
10. MeSH exp Triglycerides/
11. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
12. postprandial*.mp
13. post prandial*.mp
14. same day.mp
15. post meal*.mp
16. post cibal*.mp
17. postcibal*.mp
18. after the meal.mp
19. time course.mp
20. acute hypertriglyceridemia.mp
21. MeSH exp Postprandial Period/
22. 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21
23. endotheli*.mp
24. forearm blood flow*.mp
25. brachial arter*.mp
26. flow mediated dilat*.mp
27. flow mediated vasodilat*.mp
28. flow mediated vasoactivit*.mp
29. FMD.mp
30. MeSH exp Endothelium, Vascular/
31. MeSH exp Brachial Artery/
32. reactive hyper?emia.mp
33. 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32
34. 11 and 22 and 33
Note: exp, exploded; MeSH, Medical Subject Headings; mp, multi-purpose

Supplementary Table 2. A summary of the characteristics of the studies included in the systematic review.

	All included	NO-Dependent FMD
Study Design		
RCT (studies)	83	62
Non-RCT (studies)	48	28
Cross Over (studies)	66	49
Parallel (studies)	17	13
Participant Characteristics		
Sample Size Median (participants)	15	14
Sample Size Range (participants)	5-93	6-93
Sample Size IQR (participants)	10-20	10-20
Total participants	4061	2856
≤20 participants in study (%)	76.4	80.5
Age Average (years)	41.0	41.2
Age Range (years)	20.0-68.4	20.0-68.4
BMI Median (kg/m^2)	25.8	25.9
BMI Range (kg/m^2)	20.5-45.1	23.8-29.2
BMI IQR (kg/m^2)	23.6-28.9	21.9-45.1
Male-only (studies)	47	35
Female-only (studies)	7	5
Mixed sex (studies)	73	48
Sex not reported (studies)	4	2
Healthy (participants)	1838	1258
CVD Risk (participants)	2062	1549
CVD disease (participants)	161	49
Total CVD or risk (participants)	2223	1598
Challenge Meal Contents		
Fast Food (studies)	22	16
Cream-based (studies)	32	16
Pastry/Sandwich (studies)	31	25
Milkshake/Smoothie (studies)	13	10
Soup (studies)	5	4
Dinner meal (studies)	5	5
Breakfast meal (studies)	6	4
Not Reported (studies)	17	10

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	All included	NO-Dependent FMD
Energy Breakdown		
Total Fat (g)	66.3	64.1
Total Fat (En%)	59.5	58.5
Total Carbohydrate (g)	70.0	72.0
Total Carbohydrate (En%)	28.5	29.0
Total Protein (g)	25.5	24.0
Total Protein (En%)	10.5	10.5
Total Energy (kJ)	4197	4145
Postprandial FMD measurement time points		
1-hour (studies)	27	20
2-hour (studies)	111	85
3-hour (studies)	70	53
4-hour (studies)	136	91
5-hour (studies)	15	10
6-hour (studies)	44	24
7-hour (studies)	1	0
8-hour (studies)	18	11
FMD Analysis Method		
Manual measurement (studies)	61	33
Edge detection (studies)	70	57
Risk of Bias		
Low Risk, RCT (studies)	5	5
Some Concerns, RCT (studies)	57	51
High Risk, RCT (studies)	21	6
Low Risk, Non-RCT (studies)	0	0
Moderate Risk, Non-RCT (studies)	29	14
Serious Risk, Non-RCT (studies)	19	14
Critical Risk, Non-RCT (studies)	0	0

Abbreviations: BMI, body mass index; CVD, cardiovascular disease; En%, percentage of total meal energy; FMD, flow-mediated dilation; non-RCT, non-randomised controlled trial; RCT, randomised controlled trial.

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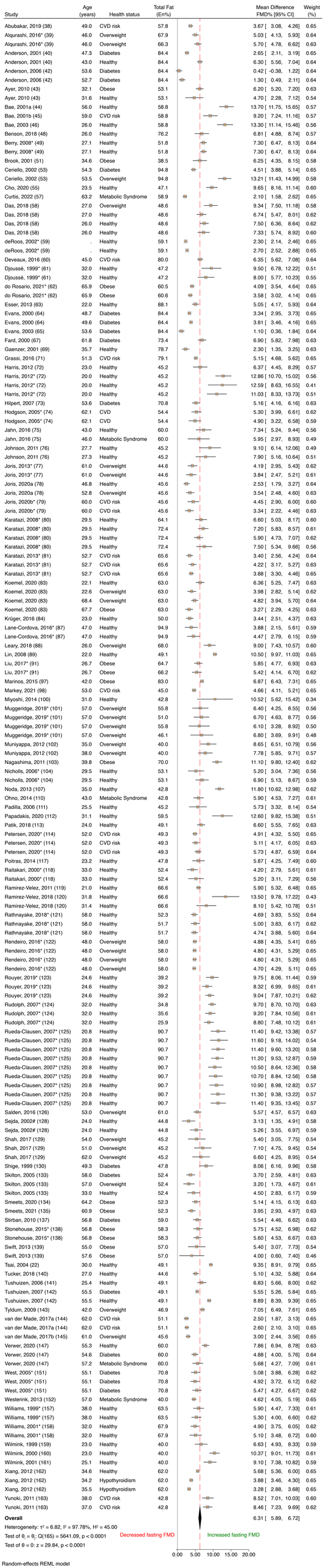
Supplementary Table 3. Other multivariable models depicting the meta-regression analysis exploring the effect of moderator covariates on FMD% effect-size variation between studies.

Covariate	Slope	Standard Error	Z-value	2-sided P-value	95% CI Lower	95% CI Upper	Obs (N)	I^2	R^2
2-hour model 2									
Intercept	-137.07	57.37	-2.39	0.017	-249.51	-24.64	79	87.02	0.409
BMI (kg/m^2)	-0.14	0.05	-3.12	0.002	-0.24	-0.05			
Fasting FMD%	-0.28	0.08	-3.63	0.000	-0.43	-0.13			
Total Energy (kJ)	0.00	0.0001	0.06	0.950	-0.00	0.00			
Total Fat (En%)	0.02	0.01	1.49	0.136	-0.01	0.04			
Sample Size (N)	0.03	0.02	1.43	0.152	-0.01	0.07			
Male N (%)	-0.00	0.00	-0.59	0.555	-0.01	0.01			
Year	0.07	0.03	2.45	0.014	0.01	0.13			
2-hour model 3									
Intercept	-138.63	56.88	-2.44	0.015	-250.11	-27.15	80	86.97	0.407
Age (years)	-0.00	0.02	-0.04	0.967	-0.03	0.03			
BMI (kg/m^2)	-0.15	0.06	-2.37	0.018	-0.27	-0.03			
Fasting FMD%	-0.28	0.08	-3.54	0.000	-0.44	-0.13			
Total Energy (kJ)	0.00	0.00	0.12	0.905	-0.00	0.00			
Total Fat (En%)	0.02	0.01	1.78	0.075	-0.00	0.04			
Sample Size (N)	0.03	0.02	1.32	0.187	-0.01	0.06			
Year	0.07	0.03	2.50	0.013	0.02	0.13			
2-hour model 4									
Intercept	-124.69	57.55	-2.17	0.030	-237.49	-11.89	79	87.42	0.382
Age (years)	0.00	0.02	0.05	0.958	-0.03	0.03			
BMI (kg/m^2)	-0.14	0.06	-2.16	0.031	-0.26	-0.01			
Fasting FMD%	-0.29	0.08	-3.57	0.000	-0.44	-0.13			
Total Energy (kJ)	0.00	0.00	0.15	0.880	-0.00	0.00			
Total Fat (En%)	0.02	0.01	1.62	0.105	-0.00	0.04			
Male N (%)	-0.00	0.00	-0.23	0.815	-0.01	0.01			
Year	0.06	0.03	2.23	0.026	0.01	0.12			
3-hour model 2									
Intercept	-93.03	55.75	-1.67	0.095	-202.31	16.24	53	64.57	0.677
Fasting FMD%	-0.27	0.07	-3.81	0.000	-0.40	-0.13			
Total Energy (kJ)	0.00	0.00	0.96	0.335	-0.00	0.00			
Total Fat (En%)	-0.03	0.01	-2.51	0.012	-0.05	-0.01			
Sample Size (N)	-0.01	0.02	-0.30	0.767	-0.05	0.04			
Year	0.05	0.03	1.71	0.087	-0.01	0.10			

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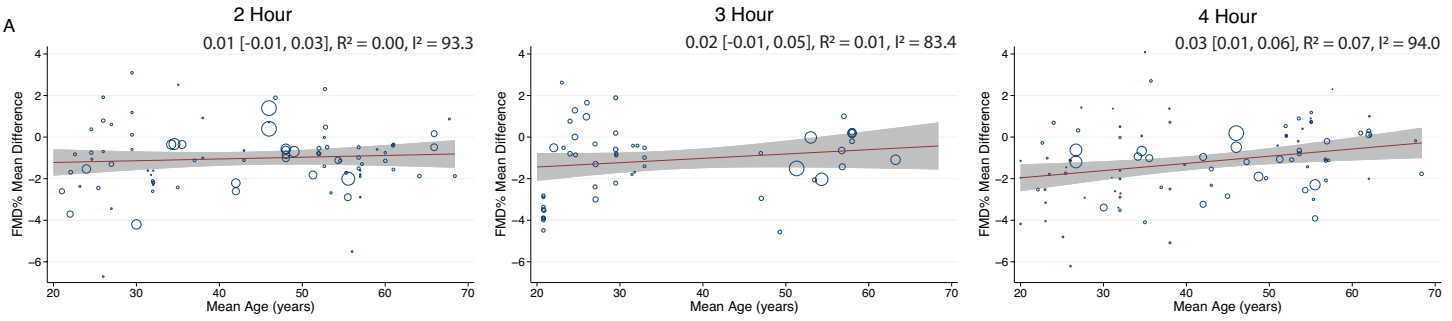
Covariate	Slope	Standard Error	Z-value	2-sided P-value	95% CI Lower	95% CI Upper	Obs (N)	I^2	R^2
4-hour model 2									
Intercept	0.15	1.55	0.09	0.925	-2.89	3.18	85	89.63	0.367
Age (years)	-0.02	0.02	-1.19	0.233	-0.05	0.01			
BMI (kg/m^2)	0.07	0.06	1.18	0.236	-0.04	0.18			
Fasting FMD%	-0.28	0.07	-4.22	0.000	-0.40	-0.15			
Total Energy (kJ)	-0.00	0.00	-2.76	0.006	-0.00	-0.00			
Total Fat (En%)	0.00	0.01	0.36	0.716	-0.02	0.03			
Sample Size (N)	0.03	0.01	2.07	0.038	0.00	0.05			
Male N (%)	-0.00	0.00	-0.28	0.778	-0.01	0.01			
4-hour model 3									
Intercept	-47.19	45.51	-1.04	0.300	-136.39	42.00	88	89.63	0.360
Age (years)	-0.01	0.01	-0.80	0.424	-0.04	0.02			
BMI (kg/m^2)	0.04	0.06	0.63	0.531	-0.08	0.15			
Fasting FMD%	-0.27	0.07	-4.19	0.000	-0.40	-0.15			
Total Energy (kJ)	-0.00	0.00	-2.06	0.039	-0.00	-0.00			
Total Fat (En%)	0.07	0.01	0.60	0.550	-0.02	0.03			
Sample Size (N)	0.02	0.01	1.72	0.085	-0.00	0.05			
Year	0.02	0.02	1.04	0.297	-0.02	0.07			
4-hour model 3									
Intercept	-61.69	47.46	-1.30	0.194	-154.72	31.33	85	90.01	0.328
Age (years)	-0.01	0.01	-0.54	0.590	-0.04	0.02			
BMI (kg/m^2)	0.05	0.06	0.84	0.399	-0.07	0.17			
Fasting FMD%	-0.28	0.07	-4.16	0.000	-0.41	-0.15			
Total Energy (kJ)	-0.00	0.00	-1.46	0.145	-0.00	0.00			
Total Fat (En%)	0.00	0.01	0.26	0.796	-0.02	0.03			
Male N (%)	-0.00	0.00	-0.35	0.727	-0.01	0.01			
Year	0.03	0.02	1.30	0.193	-0.02	0.08			

Model 1 for each timepoint can be found in the main manuscript. Random-effects meta-regression was conducted by restricted maximum likelihood. BMI indicates body mass index; CI, confidence interval; CVD, cardiovascular disease; En%, percentage of total meal energy; FMD, flow-mediated dilation; FMD%, flow-mediated dilation percent change; kJ, kilojoule; N, sample size; Obs, observations.

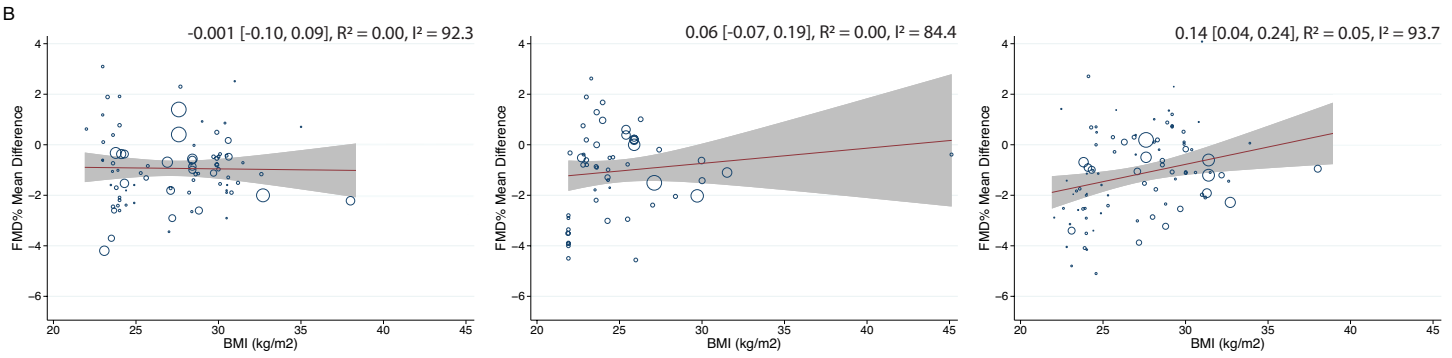

Study
Age (years)
Health status
Total Fat (En%)
Mean Difference FMD% [95% CI]
Weight (%)
Abubakar, 2019 (38) 49.0 CVD risk 57.8 3.67 [3.08, 4.26] 0.65
Alqurashi, 2016* (39) 46.0 Overweight 67.9 5.03 [4.13, 5.93] 0.64
Alqurashi, 2016* (39) 46.0 Overweight 66.3 5.70 [4.78, 6.62] 0.63
Anderson, 2001 (40) 47.3 Diabetes 84.4 2.65 [2.11, 3.19] 0.65
Anderson, 2001 (40) 43.0 Healthy 84.4 6.30 [5.56, 7.04] 0.64
Anderson, 2006 (42) 53.6 Diabetes 84.4 0.42 [-0.38, 1.22] 0.64
Anderson, 2006 (42) 52.7 Diabetes 84.4 1.30 [0.49, 2.11] 0.64
Ayer, 2010 (43) 32.1 Obese 53.1 6.20 [5.20, 7.20] 0.63
Ayer, 2010 (43) 31.6 Healthy 53.1 4.70 [2.28, 7.12] 0.54
Bae, 2001a (44) 56.0 Healthy 58.8 13.70 [11.75, 15.65] 0.57
Bae, 2001b (45) 59.0 CVD risk 58.8 9.20 [7.24, 11.16] 0.57
Bae, 2003 (46) 26.0 Healthy 58.8 13.30 [11.14, 15.46] 0.56
Benson, 2018 (48) 26.0 Healthy 76.2 6.81 [4.88, 8.74] 0.57
Berry, 2008* (49) 27.1 Healthy 51.8 7.30 [6.47, 8.13] 0.64
Berry, 2008* (49) 27.1 Healthy 51.8 7.30 [6.47, 8.13] 0.64
Brook, 2001 (51) 34.6 Obese 38.5 6.25 [4.35, 8.15] 0.58
Ceriello, 2002 (53) 54.3 Diabetes 94.8 4.51 [3.88, 5.14] 0.65
Ceriello, 2002 (53) 53.5 Overweight 94.8 13.21 [11.43, 14.99] 0.58
Cho, 2020 (55) 23.5 Healthy 47.1 9.65 [8.16, 11.14] 0.60
Curtis, 2022 (57) 63.2 Metabolic Syndrome 58.9 2.10 [1.58, 2.62] 0.65
Das, 2018 (58) 27.0 Overweight 48.6 9.34 [7.50, 11.18] 0.58
Das, 2018 (58) 27.0 Healthy 48.6 6.74 [5.47, 8.01] 0.62
Das, 2018 (58) 26.0 Healthy 48.6 7.50 [6.36, 8.64] 0.62
Das, 2018 (58) 26.0 Healthy 48.6 7.33 [5.74, 8.92] 0.60
deRoos, 2002* (59) . Healthy 59.1 2.30 [2.14, 2.46] 0.65
deRoos, 2002* (59) . Healthy 59.1 2.70 [2.52, 2.88] 0.65
Deveaux, 2016 (60) 45.0 CVD risk 80.0 6.35 [5.62, 7.08] 0.64
Djoussé, 1999* (61) 32.0 Healthy 47.2 9.50 [6.78, 12.22] 0.51
Djoussé, 1999* (61) 32.0 Healthy 47.2 8.00 [5.77, 10.23] 0.55
do Rosario, 2021* (62) 65.9 Obese 60.5 4.09 [3.54, 4.64] 0.65
do Rosario, 2021* (62) 65.9 Obese 60.6 3.58 [3.02, 4.14] 0.65
Esser, 2013 (63) 22.0 Healthy 88.1 5.05 [4.17, 5.93] 0.64
Evans, 2000 (64) 48.7 Diabetes 84.4 3.34 [2.95, 3.73] 0.65
Evans, 2000 (64) 49.6 Diabetes 84.4 3.81 [3.46, 4.16] 0.65
Evans, 2003 (65) 53.6 Diabetes 84.4 1.10 [0.36, 1.84] 0.64
Fard, 2000 (67) 61.8 Diabetes 73.4 6.90 [5.82, 7.98] 0.63
Gaenzer, 2001 (69) 35.7 Healthy 78.7 2.30 [1.35, 3.25] 0.63
Grassi, 2016 (71) 51.3 CVD risk 79.1 5.15 [4.68, 5.62] 0.65
Harris, 2012 (72) 23.0 Healthy 45.2 6.37 [4.45, 8.29] 0.57
Harris, 2012* (72) 20.0 Healthy 45.2 12.86 [10.70, 15.02] 0.56
Harris, 2012* (72) 20.0 Healthy 45.2 12.59 [8.63, 16.55] 0.41
Harris, 2012* (72) 20.0 Healthy 45.2 11.03 [8.33, 13.73] 0.51
Hilpert, 2007 (73) 53.6 Diabetes 70.8 5.16 [4.16, 6.16] 0.63
Hodgson, 2005* (74) 62.1 CVD 54.4 5.30 [3.99, 6.61] 0.62
Hodgson, 2005* (74) 62.1 CVD 54.4 4.90 [3.22, 6.58] 0.59
Jahn, 2016 (75) 43.0 Healthy 60.0 7.34 [5.24, 9.44] 0.56
Jahn, 2016 (75) 46.0 Metabolic Syndrome 60.0 5.95 [2.97, 8.93] 0.49
Johnson, 2011 (76) 27.7 Healthy 45.2 9.10 [6.14, 12.06] 0.49
Johnson, 2011 (76) 27.3 Healthy 45.2 7.90 [5.16, 10.64] 0.51
Joris, 2013* (77) 61.0 Overweight 44.6 4.19 [2.95, 5.43] 0.62
Joris, 2013* (77) 61.0 Overweight 44.6 3.84 [2.47, 5.21] 0.61
Joris, 2020a (78) 46.8 Healthy 45.6 2.53 [1.79, 3.27] 0.64
Joris, 2020a (78) 52.8 Overweight 45.6 3.54 [2.48, 4.60] 0.63
Joris, 2020b* (79) 60.0 CVD risk 45.6 4.45 [2.90, 6.00] 0.60
Joris, 2020b* (79) 60.0 CVD risk 45.6 3.34 [2.22, 4.46] 0.63
Karatzi, 2008* (80) 29.5 Healthy 64.1 6.60 [5.03, 8.17] 0.60
Karatzi, 2008* (80) 29.5 Healthy 72.4 7.20 [5.83, 8.57] 0.61
Karatzi, 2008* (80) 29.5 Healthy 72.4 5.90 [4.73, 7.07] 0.62
Karatzi, 2008* (80) 29.5 Healthy 72.4 7.50 [5.34, 9.66] 0.56
Karatzi, 2013* (81) 52.7 CVD risk 65.6 3.40 [2.56, 4.24] 0.64
Karatzi, 2013* (81) 52.7 CVD risk 65.6 4.22 [3.17, 5.27] 0.63
Karatzi, 2013* (81) 52.7 CVD risk 65.6 3.88 [3.30, 4.46] 0.65
Koemel, 2020 (83) 22.1 Healthy 63.0 6.36 [5.25, 7.47] 0.63
Koemel, 2020 (83) 22.6 Overweight 63.0 3.98 [2.82, 5.14] 0.62
Koemel, 2020 (83) 68.4 Overweight 63.0 4.82 [3.94, 5.70] 0.64
Koemel, 2020 (83) 67.7 Obese 63.0 3.27 [2.29, 4.25] 0.63
Krüger, 2016 (84) 23.0 Healthy 50.0 3.44 [2.51, 4.37] 0.63
Lane-Cordova, 2016* (87) 47.0 Healthy 94.9 3.88 [2.15, 5.61] 0.59
Lane-Cordova, 2016* (87) 47.0 Healthy 94.9 4.47 [2.79, 6.15] 0.59
Leary, 2018 (88) 26.0 Overweight 68.0 9.00 [7.43, 10.57] 0.60
Lin, 2008 (89) 22.0 Healthy 49.1 10.50 [9.97, 11.03] 0.65
Liu, 2017* (91) 26.7 Obese 64.7 5.85 [4.77, 6.93] 0.63
Liu, 2017* (91) 26.7 Obese 66.2 5.42 [4.14, 6.70] 0.62
Marinos, 2015 (97) 42.0 Obese 83.0 6.87 [6.43, 7.31] 0.65
Markey, 2021 (98) 53.0 CVD risk 45.0 4.66 [4.11, 5.21] 0.65
Miyoshi, 2014 (100) 31.0 Healthy 42.8 10.52 [5.62, 15.42] 0.34
Muggeridge, 2019* (101) 57.0 Overweight 55.8 6.40 [4.25, 8.55] 0.56
Muggeridge, 2019* (101) 57.0 Overweight 51.0 6.70 [4.63, 8.77] 0.56
Muggeridge, 2019* (101) 57.0 Overweight 55.8 6.10 [3.28, 8.92] 0.50
Muggeridge, 2019* (101) 57.0 Overweight 46.1 6.80 [3.69, 9.91] 0.48
Muniyappa, 2012 (102) 35.0 Overweight 40.0 8.65 [6.51, 10.79] 0.56
Muniyappa, 2012 (102) 38.0 Overweight 40.0 7.78 [5.85, 9.71] 0.57
Nagashima, 2011 (103) 39.8 Obese 70.0 11.10 [9.80, 12.40] 0.62
Nicholls, 2006* (104) 29.5 Healthy 53.1 5.20 [3.04, 7.36] 0.56
Nicholls, 2006* (104) 29.5 Healthy 53.1 6.90 [5.13, 8.67] 0.59
Noda, 2013 (107) 35.0 Healthy 42.8 11.80 [10.62, 12.98] 0.62
Ohno, 2014 (110) 43.0 Metabolic Syndrome 42.8 5.90 [4.53, 7.27] 0.61
Padilla, 2006 (111) 25.5 Healthy 45.2 5.73 [3.32, 8.14] 0.54
Papadakis, 2020 (112) 31.1 Healthy 59.5 12.60 [9.82, 15.38] 0.51
Patik, 2018 (113) 24.0 Healthy 49.1 6.60 [5.55, 7.65] 0.63
Petersen, 2020* (114) 52.0 CVD risk 49.3 4.91 [4.32, 5.50] 0.65
Petersen, 2020* (114) 52.0 CVD risk 49.3 5.11 [4.17, 6.05] 0.63
Petersen, 2020* (114) 52.0 CVD risk 49.3 5.73 [4.87, 6.59] 0.64
Poitras, 2014 (117) 23.2 Healthy 47.8 5.87 [4.25, 7.49] 0.60
Raitakari, 2000* (118) 33.0 Healthy 52.4 4.20 [2.79, 5.61] 0.61
Raitakari, 2000* (118) 33.0 Healthy 52.4 5.20 [3.11, 7.29] 0.56
Ramirez-Velez, 2011 (119) 21.0 Healthy 66.6 5.90 [5.32, 6.48] 0.65
Ramirez-Velez, 2018 (120) 31.8 Healthy 66.6 13.50 [9.78, 17.22] 0.43
Ramirez-Velez, 2018 (120) 31.4 Healthy 66.6 8.10 [5.42, 10.78] 0.51
Rathnayake, 2018* (121) 58.0 Healthy 52.3 4.69 [3.83, 5.55] 0.64
Rathnayake, 2018* (121) 58.0 Healthy 51.7 5.00 [3.83, 6.17] 0.62
Rathnayake, 2018* (121) 58.0 Healthy 51.7 4.74 [3.88, 5.60] 0.64
Rendeiro, 2016* (122) 48.0 Overweight 58.0 4.88 [4.35, 5.41] 0.65
Rendeiro, 2016* (122) 48.0 Overweight 58.0 4.80 [4.31, 5.29] 0.65
Rendeiro, 2016* (122) 48.0 Overweight 58.0 4.80 [4.31, 5.29] 0.65
Rendeiro, 2016* (122) 48.0 Overweight 58.0 4.70 [4.29, 5.11] 0.65
Rouyer, 2019* (123) 24.6 Healthy 39.2 9.75 [8.06, 11.44] 0.59
Rouyer, 2019* (123) 24.6 Healthy 39.2 8.32 [6.99, 9.65] 0.61
Rouyer, 2019* (123) 24.6 Healthy 39.2 9.04 [7.87, 10.21] 0.62
Rudolph, 2007* (124) 32.0 Healthy 34.8 9.70 [8.70, 10.70] 0.63
Rudolph, 2007* (124) 32.0 Healthy 35.6 9.20 [7.84, 10.56] 0.61
Rudolph, 2007* (124) 32.0 Healthy 25.9 8.80 [7.48, 10.12] 0.61
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 11.40 [9.42, 13.38] 0.57
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 11.60 [9.18, 14.02] 0.54
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 11.40 [9.60, 13.20] 0.58
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 11.20 [9.53, 12.87] 0.59
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 10.50 [8.64, 12.36] 0.58
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 10.70 [8.84, 12.56] 0.58
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 10.90 [8.98, 12.82] 0.57
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 11.30 [9.38, 13.22] 0.57
Rueda-Clausen, 2007* (125) 20.8 Healthy 90.7 11.40 [9.35, 13.45] 0.57
Salden, 2016 (126) 53.0 Overweight 61.0 5.57 [4.57, 6.57] 0.63
Sejda, 2002# (128) 24.0 Healthy 44.8 3.13 [1.35, 4.91] 0.58
Sejda, 2002# (128) 24.0 Healthy 44.8 5.26 [3.55, 6.97] 0.59
Shah, 2017 (129) 54.0 Overweight 45.2 5.40 [3.05, 7.75] 0.54
Shah, 2017 (129) 51.0 Overweight 45.2 7.10 [4.75, 9.45] 0.54
Shah, 2017 (129) 62.0 Overweight 45.2 6.60 [4.25, 8.95] 0.54
Shige, 1999 (130) 49.3 Diabetes 47.8 8.06 [6.16, 9.96] 0.58
Skilton, 2005 (133) 58.0 Diabetes 52.4 3.70 [2.59, 4.81] 0.63
Skilton, 2005 (133) 57.0 Overweight 52.4 3.20 [1.73, 4.67] 0.61
Skilton, 2005 (133) 33.0 Healthy 52.4 4.50 [2.83, 6.17] 0.59
Smeets, 2020 (134) 64.2 Obese 52.3 5.14 [4.15, 6.13] 0.63
Smeets, 2021 (135) 60.9 Obese 52.3 3.95 [2.93, 4.97] 0.63
Stirban, 2010 (137) 56.8 Diabetes 59.0 5.54 [4.46, 6.62] 0.63
Stonehouse, 2015* (138) 56.8 Obese 58.3 5.75 [4.52, 6.98] 0.62
Stonehouse, 2015* (138) 56.8 Obese 58.3 5.60 [4.53, 6.67] 0.63
Swift, 2013 (139) 55.0 Obese 57.0 5.40 [3.07, 7.73] 0.54
Swift, 2013 (139) 57.6 Obese 57.0 4.00 [0.60, 7.40] 0.46
Tsai, 2004 (22) 30.0 Healthy 49.1 9.35 [8.91, 9.79] 0.65
Tucker, 2018 (140) 27.0 Healthy 44.6 5.10 [4.32, 5.88] 0.64
Tushuizen, 2006 (141) 25.4 Healthy 49.1 6.83 [5.66, 8.00] 0.62
Tushuizen, 2007 (142) 55.5 Diabetes 49.1 5.55 [5.26, 5.84] 0.65
Tushuizen, 2007 (142) 55.5 Healthy 49.1 8.89 [8.39, 9.39] 0.65
Tyldum, 2009 (143) 42.0 Overweight 46.9 7.05 [6.49, 7.61] 0.65
van der Made, 2017a (144) 62.0 CVD risk 51.1 2.50 [1.87, 3.13] 0.65
van der Made, 2017a (144) 62.0 CVD risk 51.1 2.60 [2.10, 3.10] 0.65
van der Made, 2017b (145) 61.0 Overweight 45.6 3.00 [2.44, 3.56] 0.65
Verwer, 2020 (147) 55.3 Healthy 60.0 7.86 [6.94, 8.78] 0.63
Verwer, 2020 (147) 54.6 Diabetes 60.0 4.88 [4.00, 5.76] 0.64
Verwer, 2020 (147) 57.2 Metabolic Syndrome 60.0 5.68 [4.27, 7.09] 0.61
West, 2005* (151) 55.1 Diabetes 70.8 5.08 [3.88, 6.28] 0.62
West, 2005* (151) 55.1 Diabetes 70.8 4.92 [3.72, 6.12] 0.62
West, 2005* (151) 55.1 Diabetes 70.8 5.47 [4.27, 6.67] 0.62
Westerink, 2013 (152) 57.0 Metabolic Syndrome 40.0 4.62 [4.05, 5.19] 0.65
Williams, 1999* (157) 38.0 Healthy 63.5 5.90 [4.47, 7.33] 0.61
Williams, 1999* (157) 38.0 Healthy 63.5 5.30 [4.00, 6.60] 0.62
Williams, 2001* (158) 32.0 Healthy 67.9 4.90 [3.75, 6.05] 0.62
Williams, 2001* (158) 32.0 Healthy 67.9 5.10 [3.48, 6.72] 0.60
Wilmink, 1999 (159) 23.0 Healthy 40.0 6.63 [4.93, 8.33] 0.59
Wilmink, 2000 (160) 23.0 Healthy 40.0 10.37 [9.01, 11.73] 0.61
Wilmink, 2001 (161) 25.1 Healthy 40.0 9.10 [7.38, 10.82] 0.59
Xiang, 2012 (162) 34.6 Healthy 62.0 5.68 [5.36, 6.00] 0.65
Xiang, 2012 (162) 34.2 Hypothyroidism 62.0 3.88 [3.46, 4.30] 0.65
Xiang, 2012 (162) 35.5 Hypothyroidism 62.0 3.28 [2.88, 3.68] 0.65
Yunoki, 2011 (163) 38.0 CVD risk 42.8 8.52 [7.01, 10.03] 0.60
Yunoki, 2011 (163) 37.0 CVD risk 42.8 8.46 [7.23, 9.69] 0.62
Overall 6.31 [5.89, 6.72]
Heterogeneity: τ² = 6.82, I² = 97.78%, H² = 45.00
Test of θi = θj: Q(165) = 5641.09, p < 0.0001
Test of θ = 0: z = 29.84, p < 0.0001
Decreased fasting FMD
Increased fasting FMD
0.00
5.00
10.00
15.00
20.00
Random-effects REML model

Supplementary Figure 1. Forest plot of the mean fasting FMD%. Mean FMD% effect size and 95% CIs are indicated by white dots and black horizontal lines. The size of the boxes is proportionally scaled to the effect size for each group in the restricted maximum likelihood model. The red diamond represents the average effect size for all groups. FMD is measured as the relative percent change in peak reactive hyperemia diameter from baseline diameter (FMD%). The heterogeneity analysis is also presented. * Refers to groups with the same participants consuming different meals. # Refers to groups with the same participants consuming the same meal before and after different diet interventions. Abbreviations: CI, confidence interval; En%, percentage of total meal energy; FMD%, flow-mediated dilation percent change.

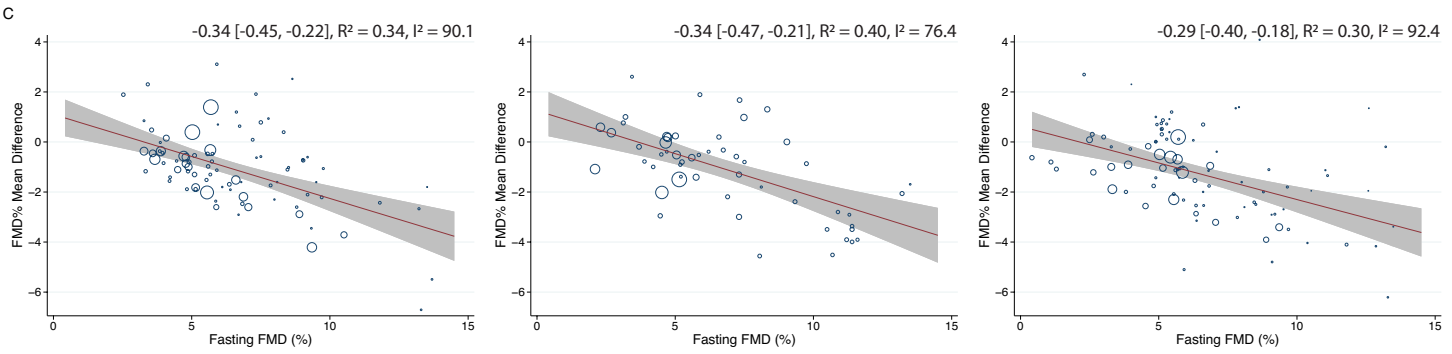
A


2 Hour
0.01 [-0.01, 0.03], R² = 0.00, I² = 93.3
3 Hour
0.02 [-0.01, 0.05], R² = 0.01, I² = 83.4
4 Hour
0.03 [0.01, 0.06], R² = 0.07, I² = 94.0
FMD% Mean Difference
Mean Age (years)

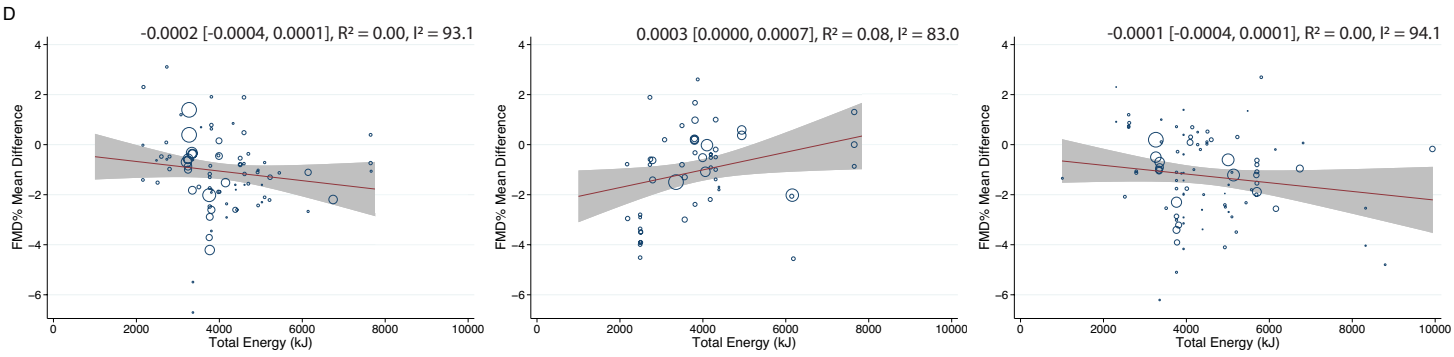
B


-0.001 [-0.10, 0.09], R² = 0.00, I² = 92.3
0.06 [-0.07, 0.19], R² = 0.00, I² = 84.4
0.14 [0.04, 0.24], R² = 0.05, I² = 93.7
FMD% Mean Difference
BMI (kg/m2)

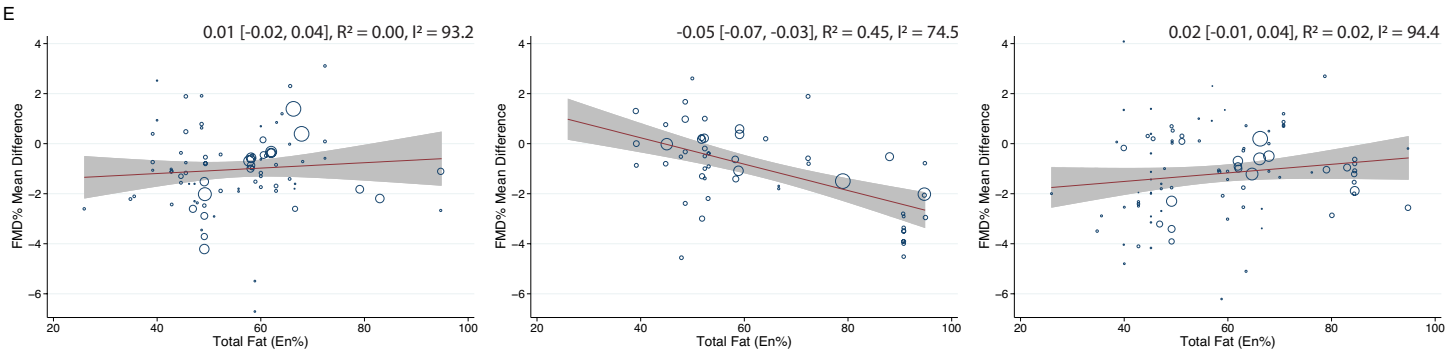
C


-0.34 [-0.45, -0.22], R² = 0.34, I² = 90.1
-0.34 [-0.47, -0.21], R² = 0.40, I² = 76.4
-0.29 [-0.40, -0.18], R² = 0.30, I² = 92.4
FMD% Mean Difference
Fasting FMD (%)

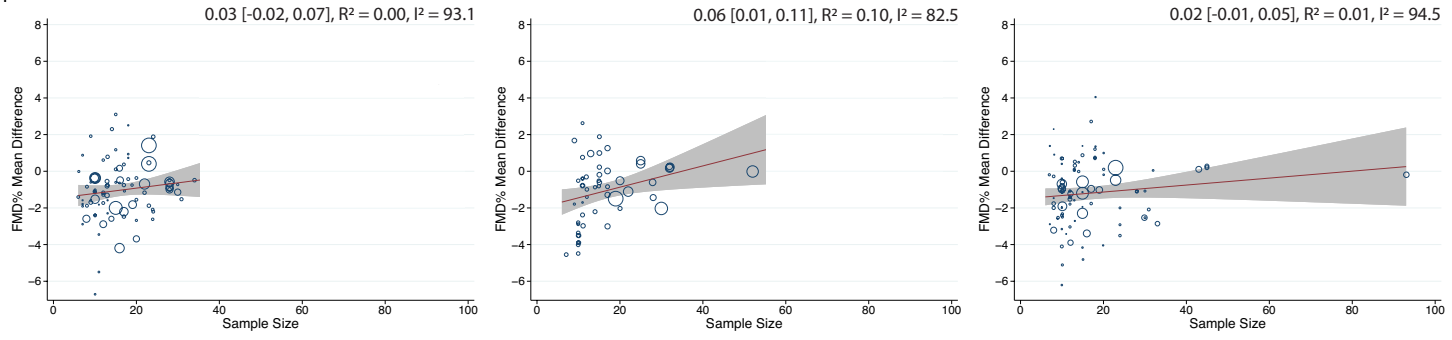
D


-0.0002 [-0.0004, 0.0001], R² = 0.00, I² = 93.1
0.0003 [0.0000, 0.0007], R² = 0.08, I² = 83.0
-0.0001 [-0.0004, 0.0001], R² = 0.00, I² = 94.1
FMD% Mean Difference
Total Energy (kJ)

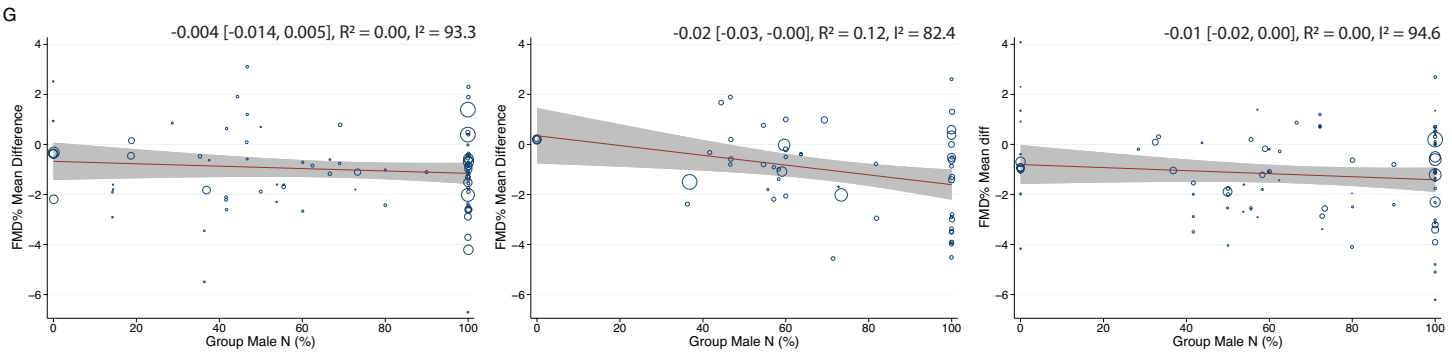
E


0.01 [-0.02, 0.04], R² = 0.00, I² = 93.2
-0.05 [-0.07, -0.03], R² = 0.45, I² = 74.5
0.02 [-0.01, 0.04], R² = 0.02, I² = 94.4
FMD% Mean Difference
Total Fat (En%)

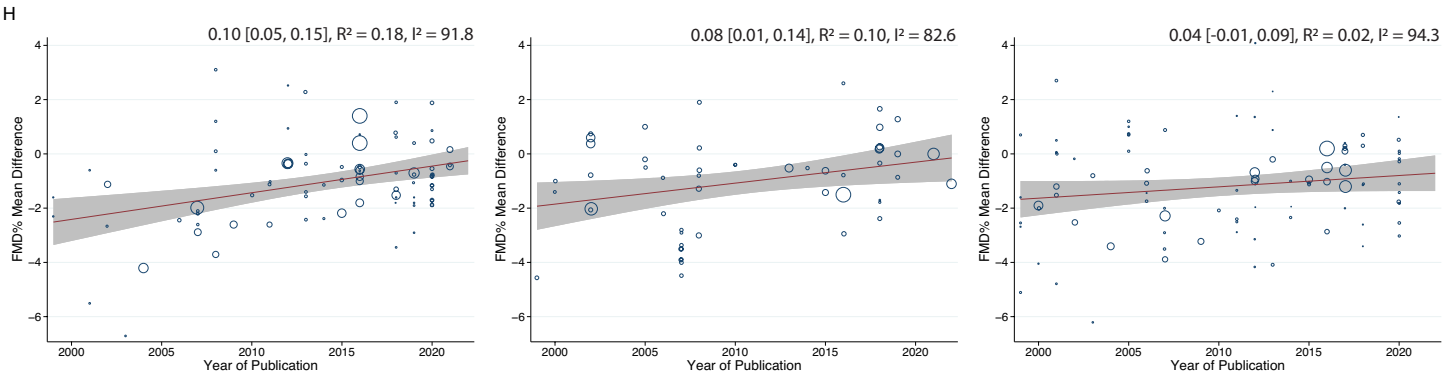
F


0.03 [-0.02, 0.07], R² = 0.00, I² = 93.1
0.06 [0.01, 0.11], R² = 0.10, I² = 82.5
0.02 [-0.01, 0.05], R² = 0.01, I² = 94.5
FMD% Mean Difference
Sample Size

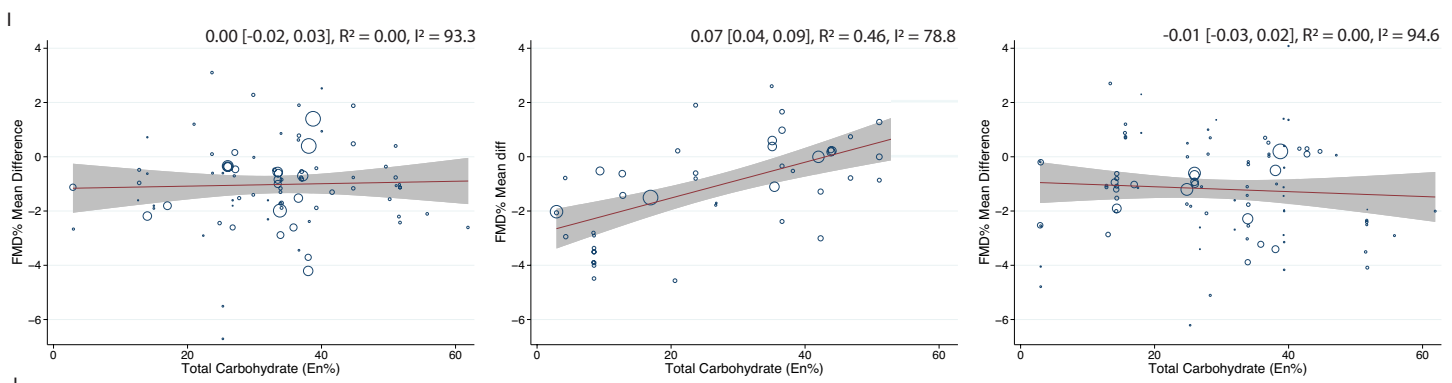
G


-0.004 [-0.014, 0.005], R² = 0.00, I² = 93.3
-0.02 [-0.03, -0.00], R² = 0.12, I² = 82.4
-0.01 [-0.02, 0.00], R² = 0.00, I² = 94.6
FMD% Mean Difference
FMD% Mean diff
Group Male N (%)

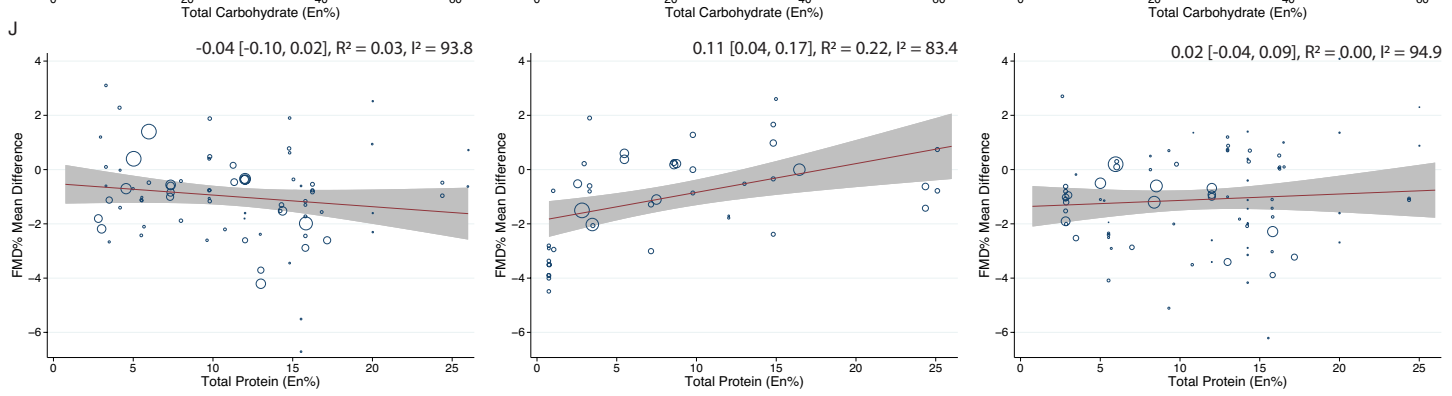
H


0.10 [0.05, 0.15], R² = 0.18, I² = 91.8
0.08 [0.01, 0.14], R² = 0.10, I² = 82.6
0.04 [-0.01, 0.09], R² = 0.02, I² = 94.3
FMD% Mean Difference
Year of Publication

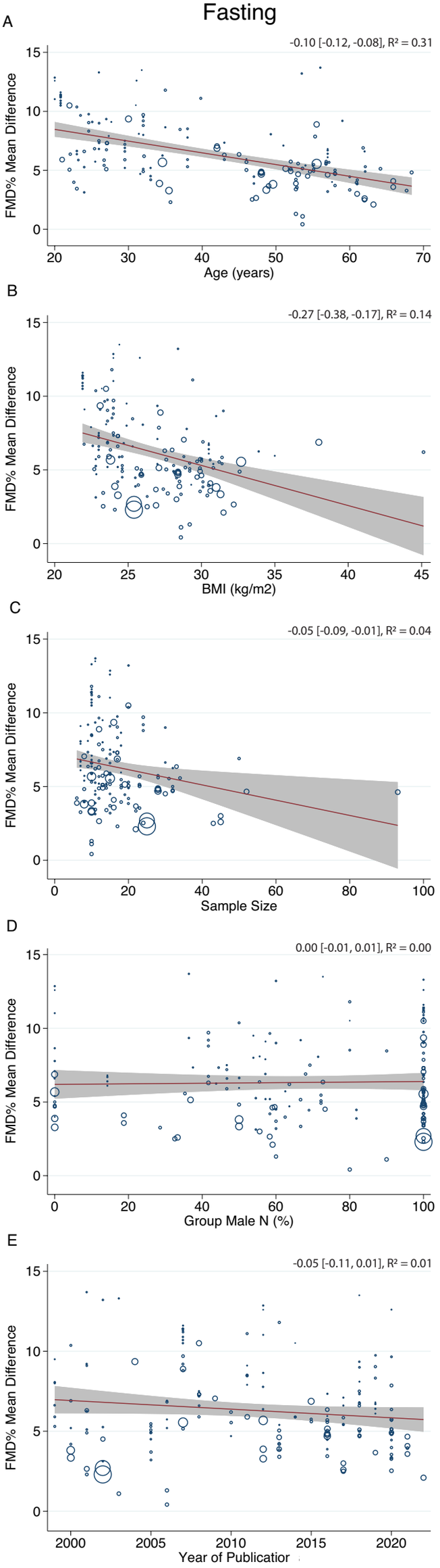
I


0.00 [-0.02, 0.03], R² = 0.00, I² = 93.3
0.07 [0.04, 0.09], R² = 0.46, I² = 78.8
-0.01 [-0.03, 0.02], R² = 0.00, I² = 94.6
FMD% Mean Difference
FMD% Mean diff
Total Carbohydrate (En%)

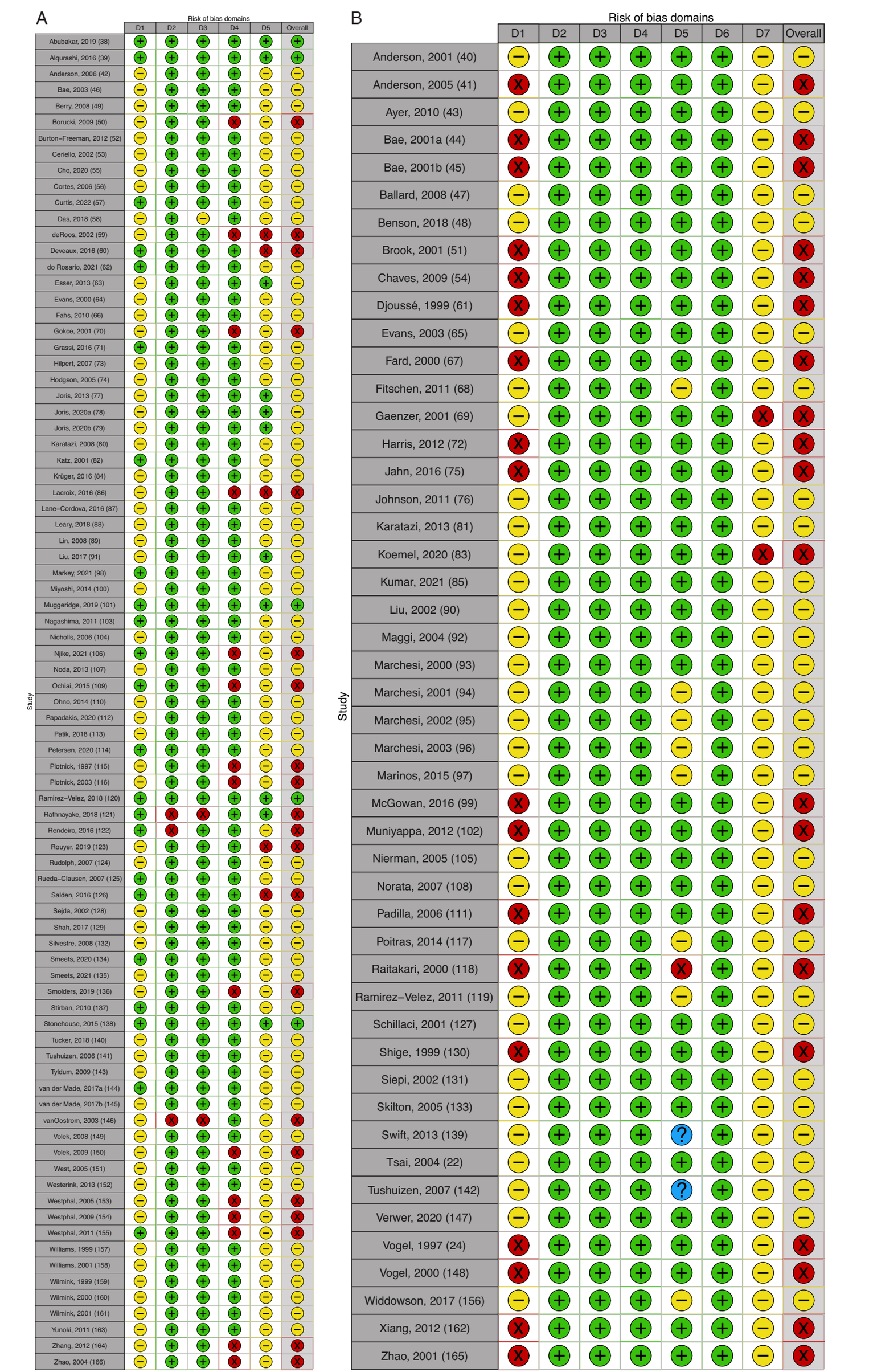
J


-0.04 [-0.10, 0.02], R² = 0.03, I² = 93.8
0.11 [0.04, 0.17], R² = 0.22, I² = 83.4
0.02 [-0.04, 0.09], R² = 0.00, I² = 94.9
FMD% Mean Difference
Total Protein (En%)

Supplementary Figure 2. Unadjusted linear regression analysis to evaluate the association of potential modifiers A) age, B) BMI, C) fasting FMD%, D) total meal energy, E) meal fat percentage, F) sample size, G) percentage of male participants, H) year of publication, I) meal carbohydrate percentage, and J) meal protein percentage on the effect of a single high-fat meal on endothelial function. Bubble plot with linear trend line (red line) and 95% CI (grey area). FMD is measured as the relative percent change in peak reactive hyperemia diameter from baseline diameter (FMD%). The mean difference in FMD percentage change was calculated as fasting FMD% subtracted from postprandial FMD%, termed FMD change; the units of which are percentage points (pp). Bubbles are the inverse variance weighted mean difference in FMD% (postprandial-fasting). Coefficient [CI; Lower, Upper], and R^2 are displayed in the top right-hand corner. Abbreviations: CI, confidence interval; En%, percentage of total meal energy; FMD%, flow-mediated dilation percent change; N, sample size.


Fasting
A
-0.10 [-0.12, -0.08], R² = 0.31
FMD% Mean Difference
Age (years)
B
-0.27 [-0.38, -0.17], R² = 0.14
FMD% Mean Difference
BMI (kg/m2)
C
-0.05 [-0.09, -0.01], R² = 0.04
FMD% Mean Difference
Sample Size
D
0.00 [-0.01, 0.01], R² = 0.00
FMD% Mean Difference
Group Male N (%)
E
-0.05 [-0.11, 0.01], R² = 0.01
FMD% Mean Difference
Year of Publicatior

Supplementary Figure 3. Unadjusted linear regression analysis to evaluate the association of potential modifiers A) age, B) BMI, C) sample size, D) percentage of male participants, and E) year of publication on fasting FMD%. Bubble plot with linear trend line (red line) and 95% CI (grey area). FMD is measured as the relative percent change in peak reactive hyperemia diameter from baseline diameter (FMD%). The mean difference in FMD percentage change was calculated as fasting FMD% subtracted from postprandial FMD%, termed FMD change; the units of which are percentage points (pp). Bubbles are the inverse variance weighted mean effect size in fasting FMD%. Coefficient [CI: Lower, Upper], and R^2 are displayed in the top right-hand corner. Abbreviations: BMI, body mass index; CI, confidence interval; En%, percentage of total meal energy; FMD%, flow-mediated dilation percent change; N, sample size.


A
Risk of bias domains
D1
D2
D3
D4
D5
Overall
Study
Abubakar, 2019 (38)
Alqurashi, 2016 (39)
Anderson, 2006 (42)
Bae, 2003 (46)
Berry, 2008 (49)
Borucki, 2009 (50)
Burton−Freeman, 2012 (52)
Ceriello, 2002 (53)
Cho, 2020 (55)
Cortes, 2006 (56)
Curtis, 2022 (57)
Das, 2018 (58)
deRoos, 2002 (59)
Deveaux, 2016 (60)
do Rosario, 2021 (62)
Esser, 2013 (63)
Evans, 2000 (64)
Fahs, 2010 (66)
Gokce, 2001 (70)
Grassi, 2016 (71)
Hilpert, 2007 (73)
Hodgson, 2005 (74)
Joris, 2013 (77)
Joris, 2020a (78)
Joris, 2020b (79)
Karatazi, 2008 (80)
Katz, 2001 (82)
Krüger, 2016 (84)
Lacroix, 2016 (86)
Lane−Cordova, 2016 (87)
Leary, 2018 (88)
Lin, 2008 (89)
Liu, 2017 (91)
Markey, 2021 (98)
Miyoshi, 2014 (100)
Muggeridge, 2019 (101)
Nagashima, 2011 (103)
Nicholls, 2006 (104)
Njike, 2021 (106)
Noda, 2013 (107)
Ochiai, 2015 (109)
Ohno, 2014 (110)
Papadakis, 2020 (112)
Patik, 2018 (113)
Petersen, 2020 (114)
Plotnick, 1997 (115)
Plotnick, 2003 (116)
Ramirez−Velez, 2018 (120)
Rathnayake, 2018 (121)
Rendeiro, 2016 (122)
Rouyer, 2019 (123)
Rudolph, 2007 (124)
Rueda−Clausen, 2007 (125)
Salden, 2016 (126)
Sejda, 2002 (128)
Shah, 2017 (129)
Silvestre, 2008 (132)
Smeets, 2020 (134)
Smeets, 2021 (135)
Smolders, 2019 (136)
Stirban, 2010 (137)
Stonehouse, 2015 (138)
Tucker, 2018 (140)
Tushuizen, 2006 (141)
Tyldum, 2009 (143)
van der Made, 2017a (144)
van der Made, 2017b (145)
vanOostrom, 2003 (146)
Volek, 2008 (149)
Volek, 2009 (150)
West, 2005 (151)
Westerink, 2013 (152)
Westphal, 2005 (153)
Westphal, 2009 (154)
Westphal, 2011 (155)
Williams, 1999 (157)
Williams, 2001 (158)
Wilmink, 1999 (159)
Wilmink, 2000 (160)
Wilmink, 2001 (161)
Yunoki, 2011 (163)
Zhang, 2012 (164)
Zhao, 2004 (166)
B
Risk of bias domains
D1
D2
D3
D4
D5
D6
D7
Overall
Study
Anderson, 2001 (40)
Anderson, 2005 (41)
Ayer, 2010 (43)
Bae, 2001a (44)
Bae, 2001b (45)
Ballard, 2008 (47)
Benson, 2018 (48)
Brook, 2001 (51)
Chaves, 2009 (54)
Djoussé, 1999 (61)
Evans, 2003 (65)
Fard, 2000 (67)
Fitschen, 2011 (68)
Gaenzer, 2001 (69)
Harris, 2012 (72)
Jahn, 2016 (75)
Johnson, 2011 (76)
Karatazi, 2013 (81)
Koemel, 2020 (83)
Kumar, 2021 (85)
Liu, 2002 (90)
Maggi, 2004 (92)
Marchesi, 2000 (93)
Marchesi, 2001 (94)
Marchesi, 2002 (95)
Marchesi, 2003 (96)
Marinos, 2015 (97)
McGowan, 2016 (99)
Muniyappa, 2012 (102)
Nierman, 2005 (105)
Norata, 2007 (108)
Padilla, 2006 (111)
Poitras, 2014 (117)
Raitakari, 2000 (118)
Ramirez−Velez, 2011 (119)
Schillaci, 2001 (127)
Shige, 1999 (130)
Siepi, 2002 (131)
Skilton, 2005 (133)
Swift, 2013 (139)
Tsai, 2004 (22)
Tushuizen, 2007 (142)
Verwer, 2020 (147)
Vogel, 1997 (24)
Vogel, 2000 (148)
Widdowson, 2017 (156)
Xiang, 2012 (162)
Zhao, 2001 (165)

Supplementary Figure 4. Risk of bias assessment of included A) RCT and B) Non-RCT articles by Cochrane Risk of Bias 2.0 (ROB2.0) and Risk Of Bias In Non-randomized Studies - of Interventions (ROBINS-I) tool respectively.

For ROB2.0, D1: Bias arising from the randomization process; D2: Bias due to deviations from intended interventions; D3: Bias due to missing outcome data; D4: Bias in measurement of the outcome; D5: Bias in selection of the reported result.

For ROBINS-I, D1: Bias due to confounding; D2: Bias due to selection of participants; D3: Bias in classification of interventions; D4: Bias due to deviations from intended interventions; D5 Bias due to missing data; D6 Bias in measurement of outcomes; D7: Bias in the selection of the reported result.

Green plus symbol denotes low risk of bias, yellow minus symbol denotes moderate risk of bias, red cross symbol denotes high risk of bias, and blue question mark symbol denotes no information.

Created using *robvis.*

McGuinness, LA, Higgins, JPT. Risk-of-bias VISualization (robvis): An R package and Shiny web app for visualizing risk-of-bias assessments. Res Syn Meth. 2020; 1- 7. https://doi.org/10.1002/jrsm.1411